

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 9: Due by 11:59pm on Sunday April 21, 2024.

Submit to WyoCourses a Jupyter notebook (both ipynb and HTML files) .
Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [60 points] Consider the heat equation $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ with initial-boundary conditions

$$u(x, 0) = e^{-x/2} \quad \text{for } x \in [0, 1]$$

$$u(0, t) = e^t \quad \text{for } t \in [0, 1]$$

$$u(1, t) = e^{t-1/2} \quad \text{for } t \in [0, 1].$$

- (A) [10 points] Prove that $u(x, t) = e^{t-x/2}$ is a solution and plot $u(x, t)$ for $x \in [0, 1]$ and $t \in [0, 1]$. I recommend plotting 5 curves for times $t_j \in \{0, 0.1, 0.2, 0.5, 1\}$.
- (B) [10 points] Approximately solve the above heat equation using the Forward Difference Method with spatial step size $h = 0.1$ and time step $k = 0.001$. In two plots, show the approximate solution u_{ij} as well as the error $e_{ij} = |u(x_i, t_j) - u_{i,j}|$ for $i \in \{0, \dots, n\}$ and $j \in \{0, \dots, N\}$. (I recommend plotting the true solution as 5 solid lines and the approximations as symbols.)
- (C) [10 points] Repeat problem 1B, but use the Backward Difference Method.
- (D) [10 points] Repeat problem 1B, but use the Crank-Nicolson Method.
- (E) [15 points] Study how the approximation error changes one you fix $h = 0.1$ and vary the time step $k \in \{0.01, 0.005, 0.001, 0.0005\}$ (i.e., $N \in \{100, 200, 1000, 2000\}$). Focus on the position $x = 0.5$ and $t = 1$ and study the error $e_{i,j} = |u(0.5, 1) - u_{i,j}|$ versus k . (Note that as you vary k , the i and j values corresponding to $(x, t) = (0.5, 1)$ with change.) Plot 3 lines for the errors at $(x, t) = (0.5, 1)$ for the 3 methods: Forward Difference, Backward Difference and Crank-Nicolson.
- (F) [5 points] Which of the simulations in Problem 1E are Von Neumann stable and why?

2. [40 points] Consider the wave equation $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$ (i.e., $c = 2$) with initial-boundary conditions

$$u(x, 0) = 0 \quad \text{for } x \in [0, 1]$$

$$u_t(x, 0) = 2\pi \sin(\pi x) \quad \text{for } x \in [0, 1]$$

$$u(0, t) = 0 \quad \text{for } t \in [0, 1]$$

$$u(1, t) = 0 \quad \text{for } t \in [0, 1].$$

- (A) [10 points] Prove that $u(x, t) = \sin(\pi x) \sin(2\pi t)$ is a solution and plot $u(x, t)$ for $x \in [0, 1]$ and $t \in [0, 1]$. (I suggest a surface plot here and in Problem 2B.)
- (B) [15 points] Approximately solve the above wave equation using the finite difference method with spatial step size $h = 0.05$ and time step $k = h/2$. In two plots, show the approximate solution u_{ij} as well as the error $e_{ij} = |u(x_i, t_j) - u_{i,j}|$ for $i \in \{0, \dots, n\}$ and $j \in \{0, \dots, N\}$.
- (C) [15 points] Study how the approximation error changes one you vary the spatial and temporal step sizes $h = 2^{-p}$ with $p \in \{4, 5, 6, 7, 8\}$ and $k = h/2$. Focus on the position $x = 1/4$ and $t = 3/4$ and study the error $e_{i,j} = |u(1/4, 3/4) - u_{i,j}|$ versus p . (Note that as you vary p , the i and j values corresponding to $(x, t) = (1/4, 3/4)$ with change.)